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Data Structures and Algorithms – Midterm Examination

Question 1 (5 Marks):

What is the time complexity of Binary Search? Explain briefly.

Answer:

The time complexity of Binary Search is $O(\log n)$.

Binary Search can only be used on a sorted array. It checks the middle element and compares it with the target value.

If the target is smaller, it searches the left half. If the target is larger, it searches the right half.

Since half of the remaining elements are discarded in every step, the number of comparisons grows logarithmically.